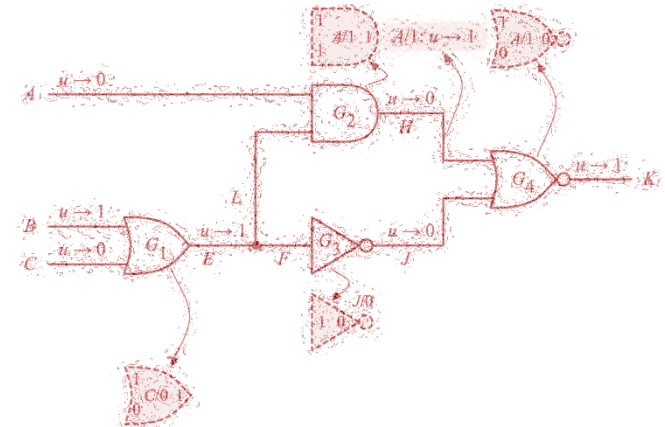


Fault Simulation

- Introduction
- Fault simulation techniques
 - ◆ Serial fault simulation
 - ◆ Parallel fault simulation (1965)
 - ◆ PPSFP (1985)
 - ◆ Deductive fault simulation (1972)
 - ◆ Concurrent fault simulation (1974)
 - ◆ Differential fault simulation (1989)
- Alternatives to fault simulation
- Issues of fault simulation
- Concluding remarks



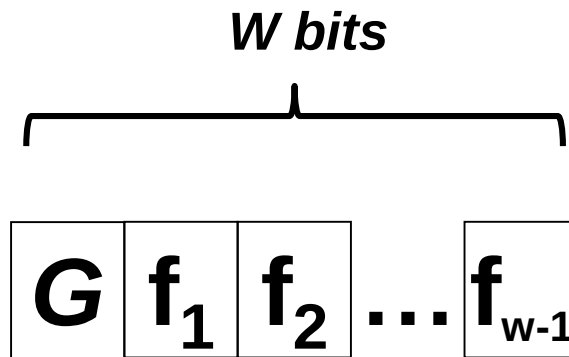
Two Types of Parallelism

- Exploit parallelism of logic instructions (bit-wise operations)
 - ◆ 1. Parallelism in faults
 - * *Parallel Fault Simulation* [Seshu 65]
 - ◆ 2. Parallelism in patterns
 - * *Parallel Pattern Single Fault Propagation, PPSFP* [Wacukauski 85]

Two Types of Parallelism

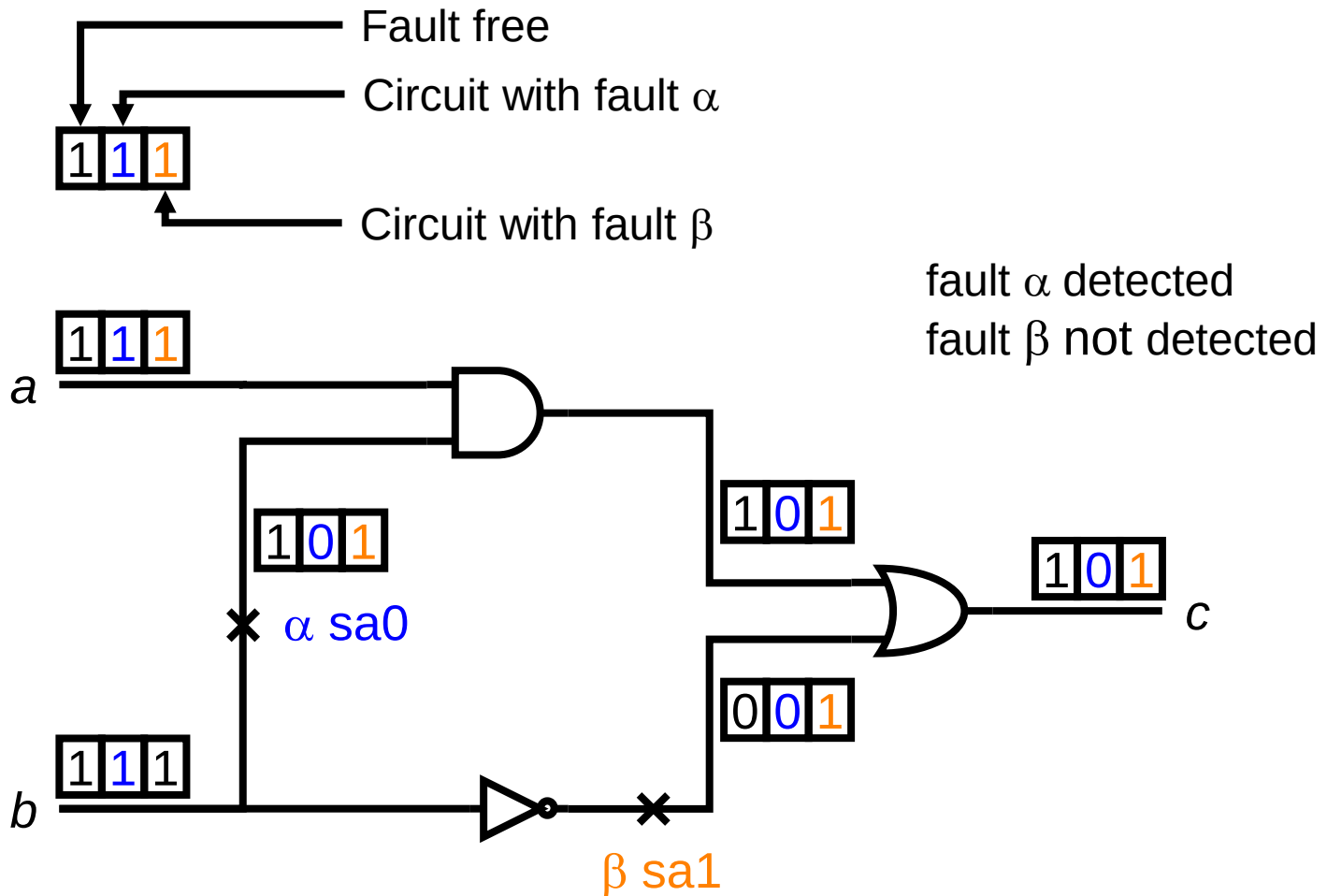
Parallel Fault Simulation [Seshu 65]

- Run parallel logic simulation with $W-1$ faults (W is CPU word size)
 - ♦ one bit for **fault-free** circuit
 - ♦ $W-1$ bits for **faulty** circuit
- ($W-1$) times faster than serial fault simulation (w/o fault dropping)



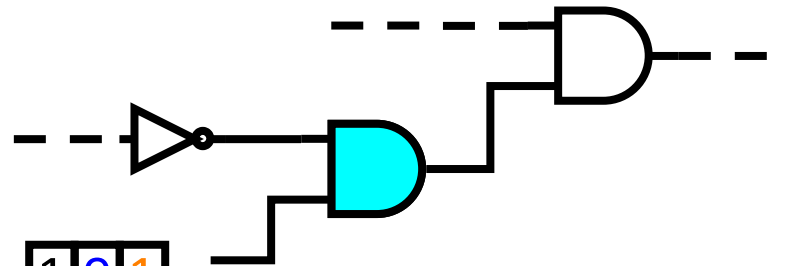
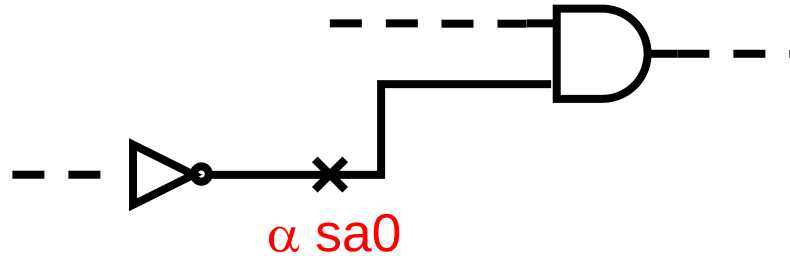
Example

- Assume $W=3$



Fault Injection

- **Fault Injection** = inserting fault(s) into circuit



Control input

1	0	1
---	---	---

0 = SA0 fault injected

1 = SA0 fault not injected

Quiz: how to insert SA1 fault?

A:

Quiz

Q: Consider two faults: $(f=SA1, g=SA0)$. Use parallel fault simulation on patterns

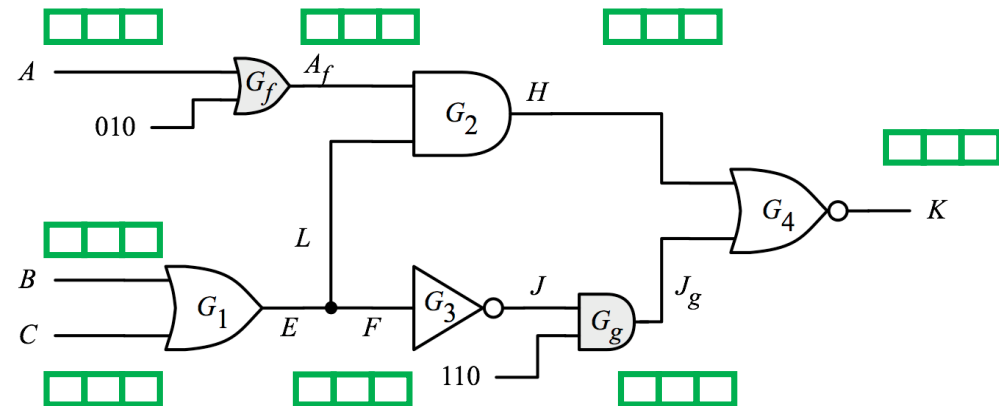
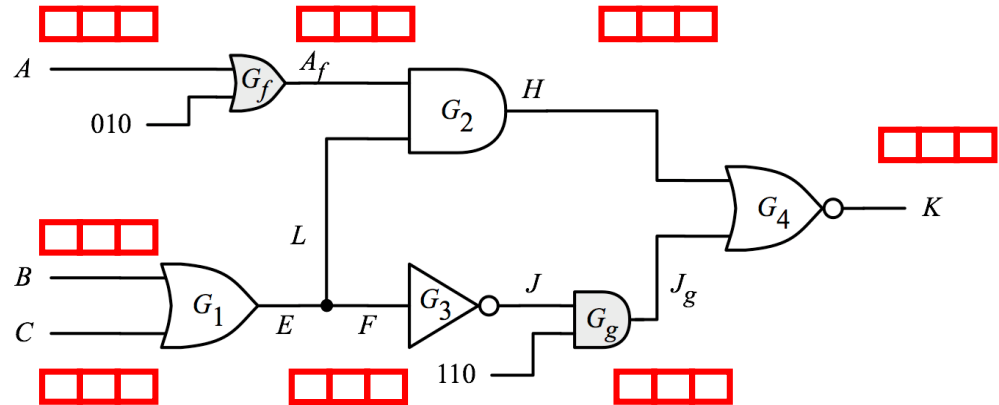
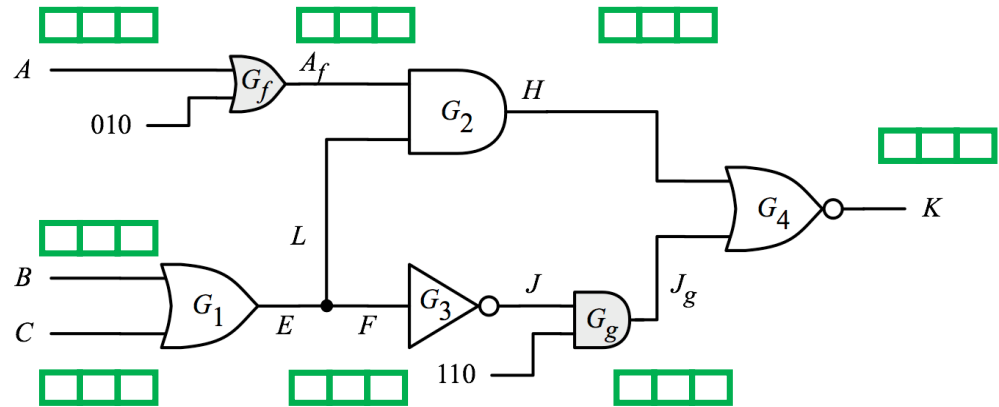
$P_1 = \{010\}$,

$P_2 = \{001\}$,

$P_3 = \{100\}$.

A:

Assume no fault dropping



Solution

Q: Consider two faults: ($f=SA1$, $g=SA0$). Use parallel fault simulation on patterns

$P_1 = \{010\}$,

$P_2 = \{001\}$,

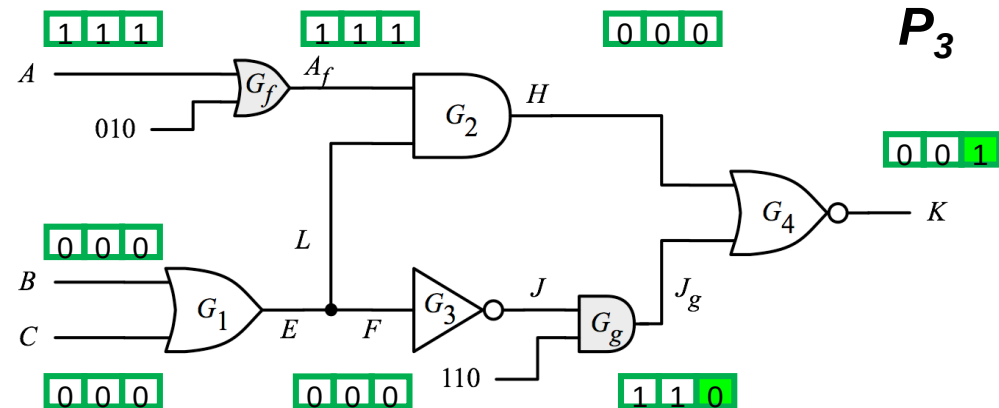
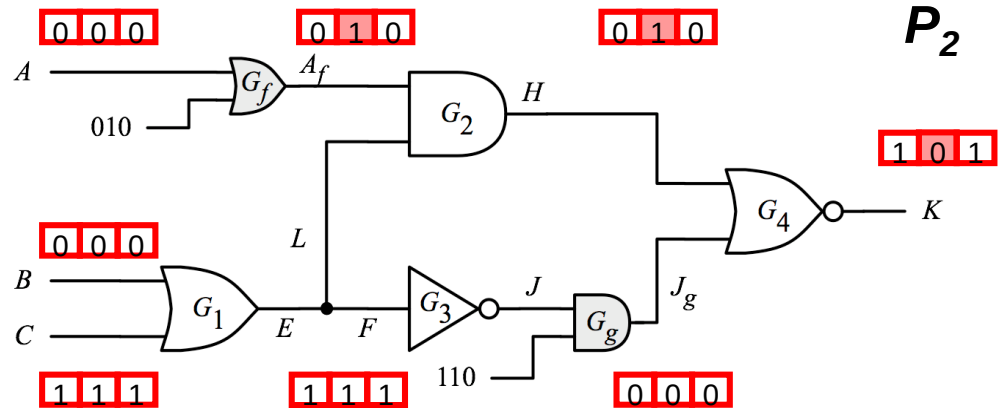
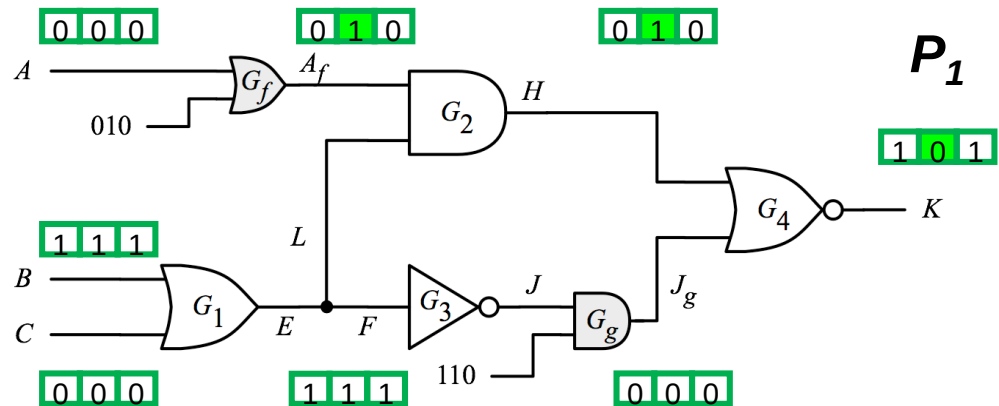
$P_3 = \{100\}$.

A:

$P_1 P_2$ detect f

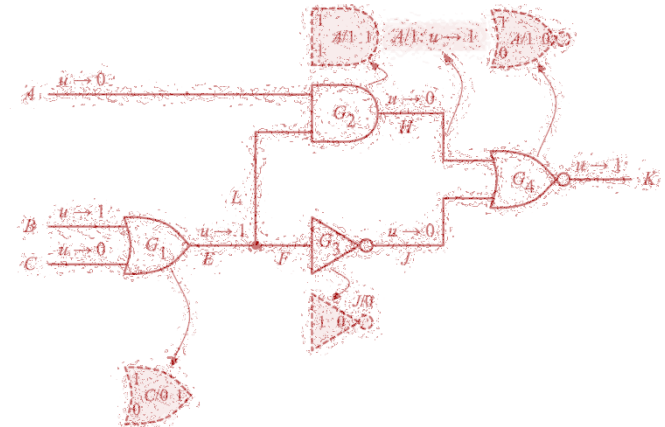
P_3 detects g

Assume no fault dropping



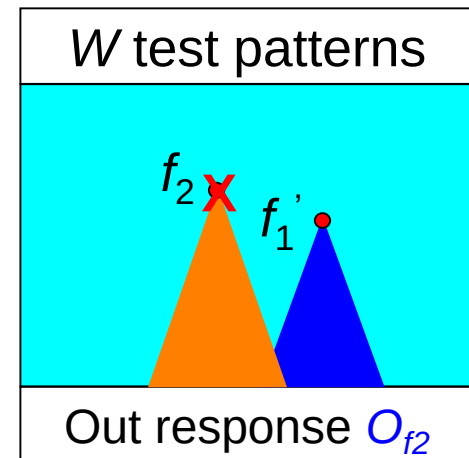
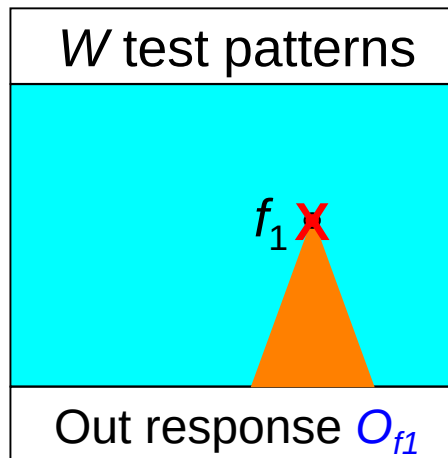
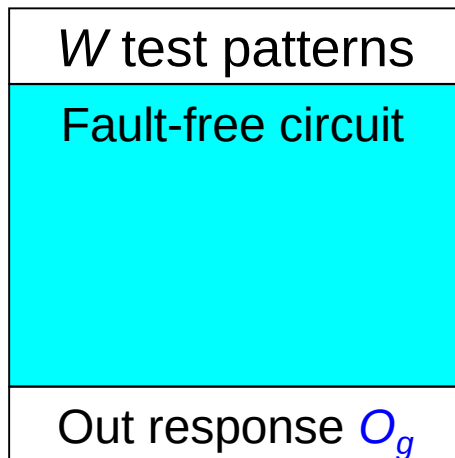
Fault Simulation

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PPSFP

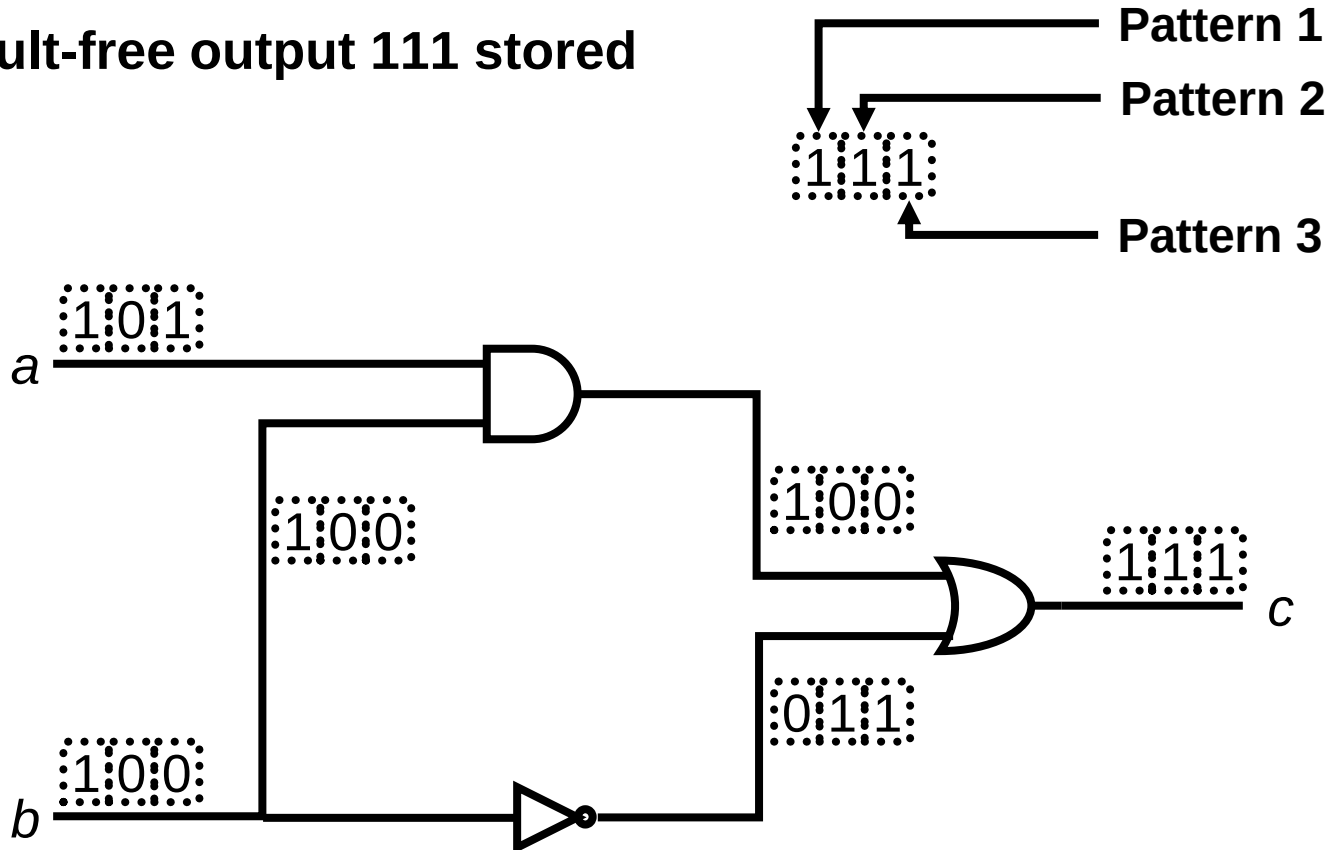
- Parallel pattern assumes **combinational circuit**
- Procedure
 - ◆ Step 0: run logic simulation, store good output O_g
 - ◆ Step 1: Inject f_1 by creating an event at the fault site
 - ◆ Step 2: Perform **event-driven simulation**, get output O_{f1}
 - * Compare O_g and O_{f1}
 - ◆ Step 3. Create an event to **undo** fault effect of f_1
 - * Inject an event for next fault f_2
 - ◆ Repeat step 2~3 until all faults simulated



...

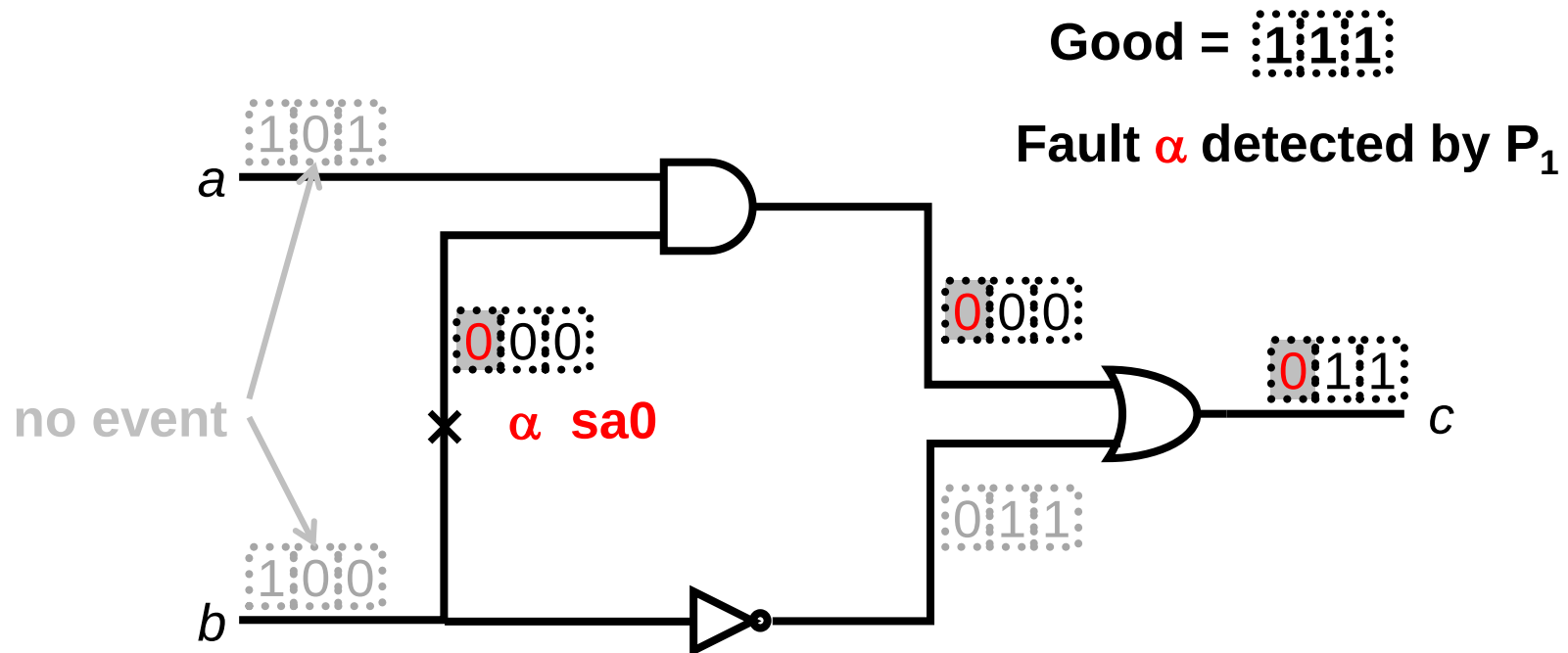
PPSFP Example

- Assume $W=3$
- Fault-free circuit
 - ♦ Fault-free output 111 stored



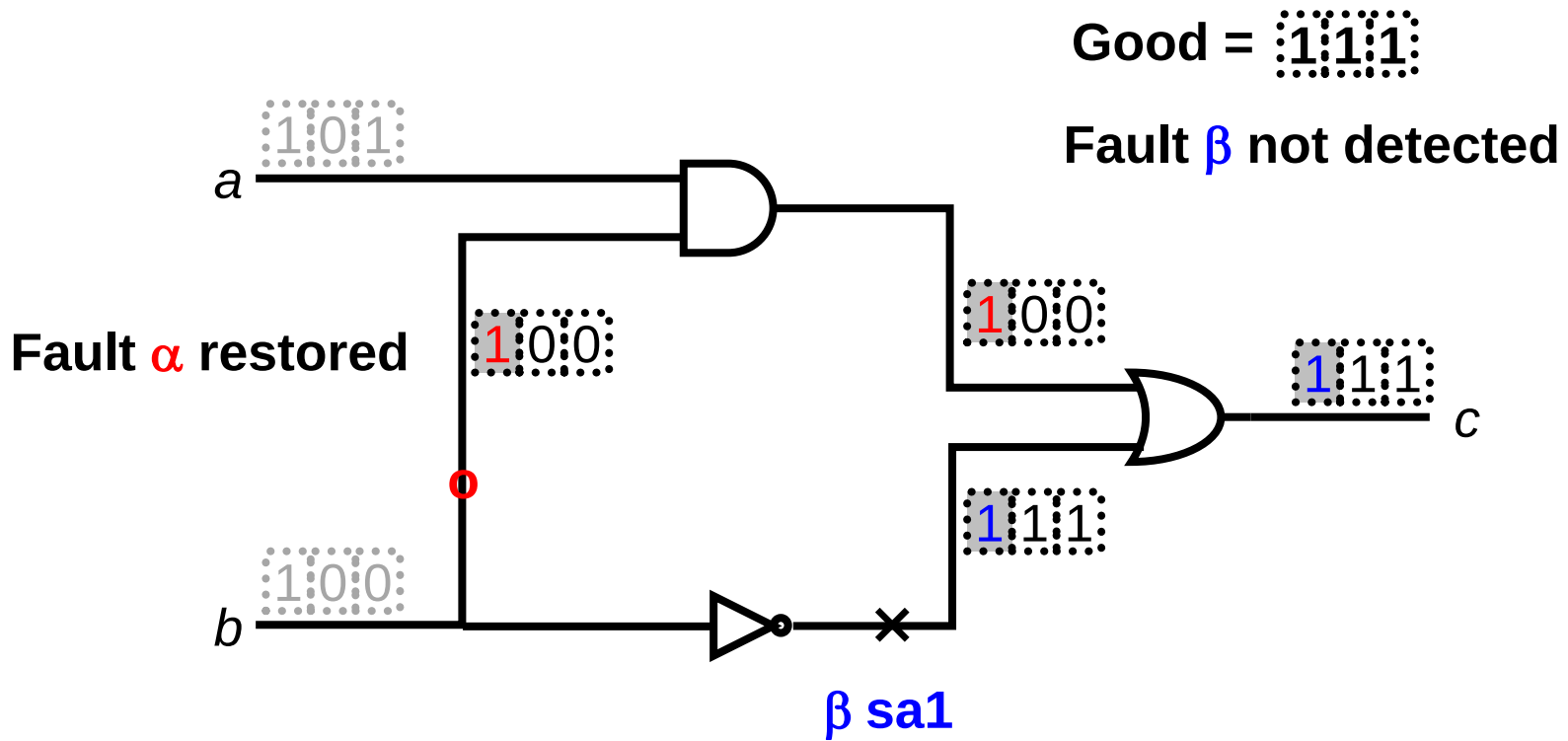
Example- cont'd

- One event created to inject fault α
 - ♦ fault effects are then propagated



Example – cont'd

- Two events created:
 - One event to restore fault α
 - One event to inject fault β

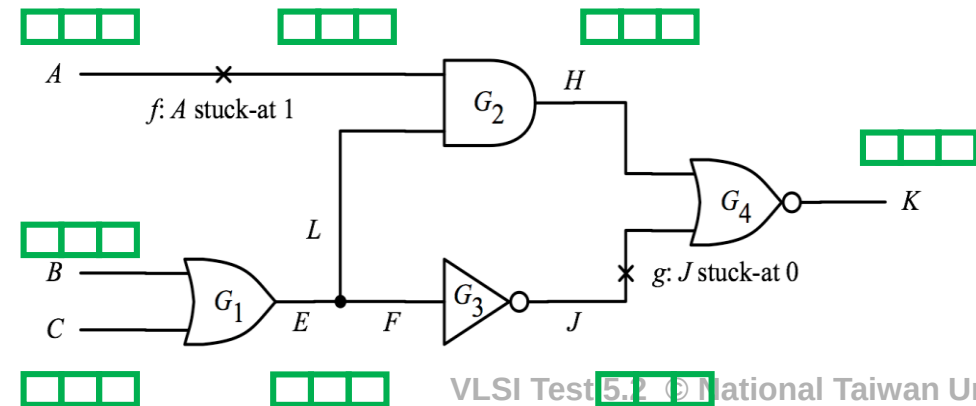
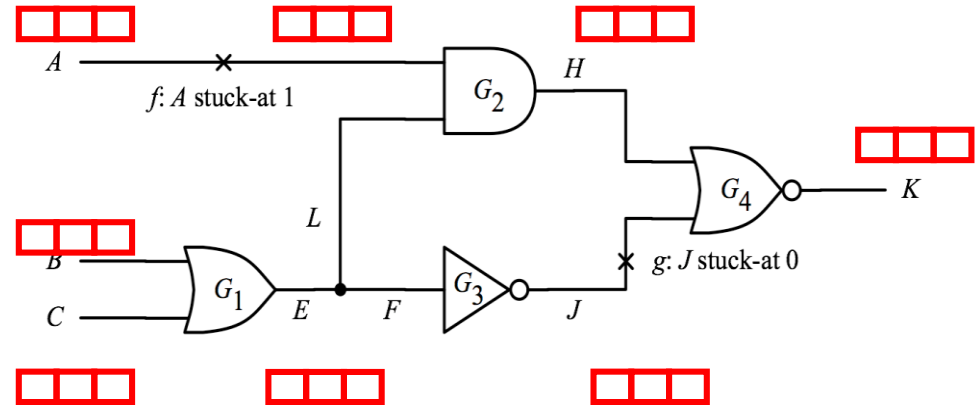
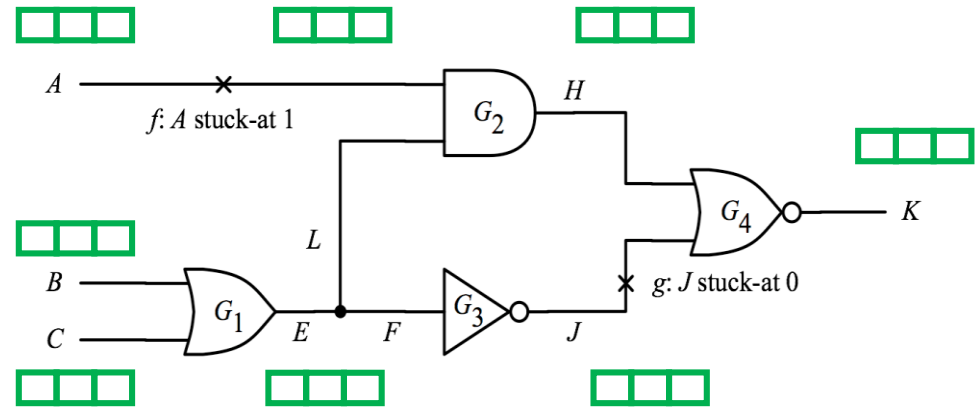


Quiz

Q: Consider two faults: ($f=SA1$, $g=SA0$). Use PPSFP on patterns

$$P_1 = \{010\},$$
$$P_2 = \{001\},$$
$$P_3 = \{100\}.$$

A:



Solution

Q: Consider two faults: ($f=SA1$, $g=SA0$). Use PPSFP on patterns

$P_1 = \{010\}$,

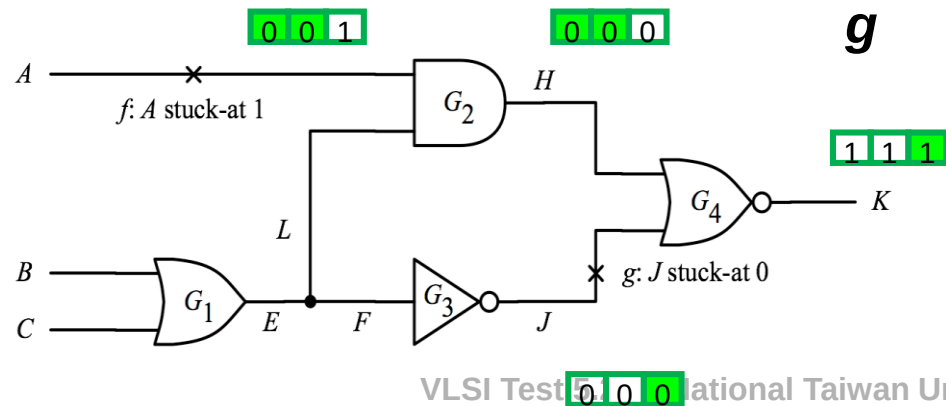
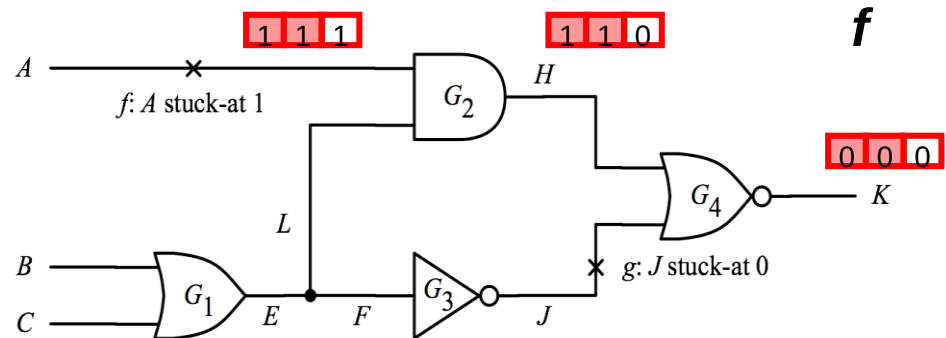
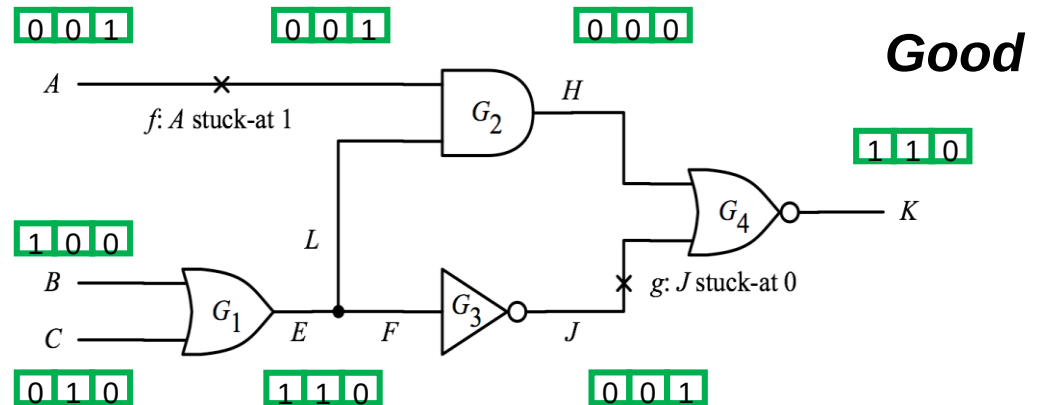
$P_2 = \{001\}$,

$P_3 = \{100\}$.

A:

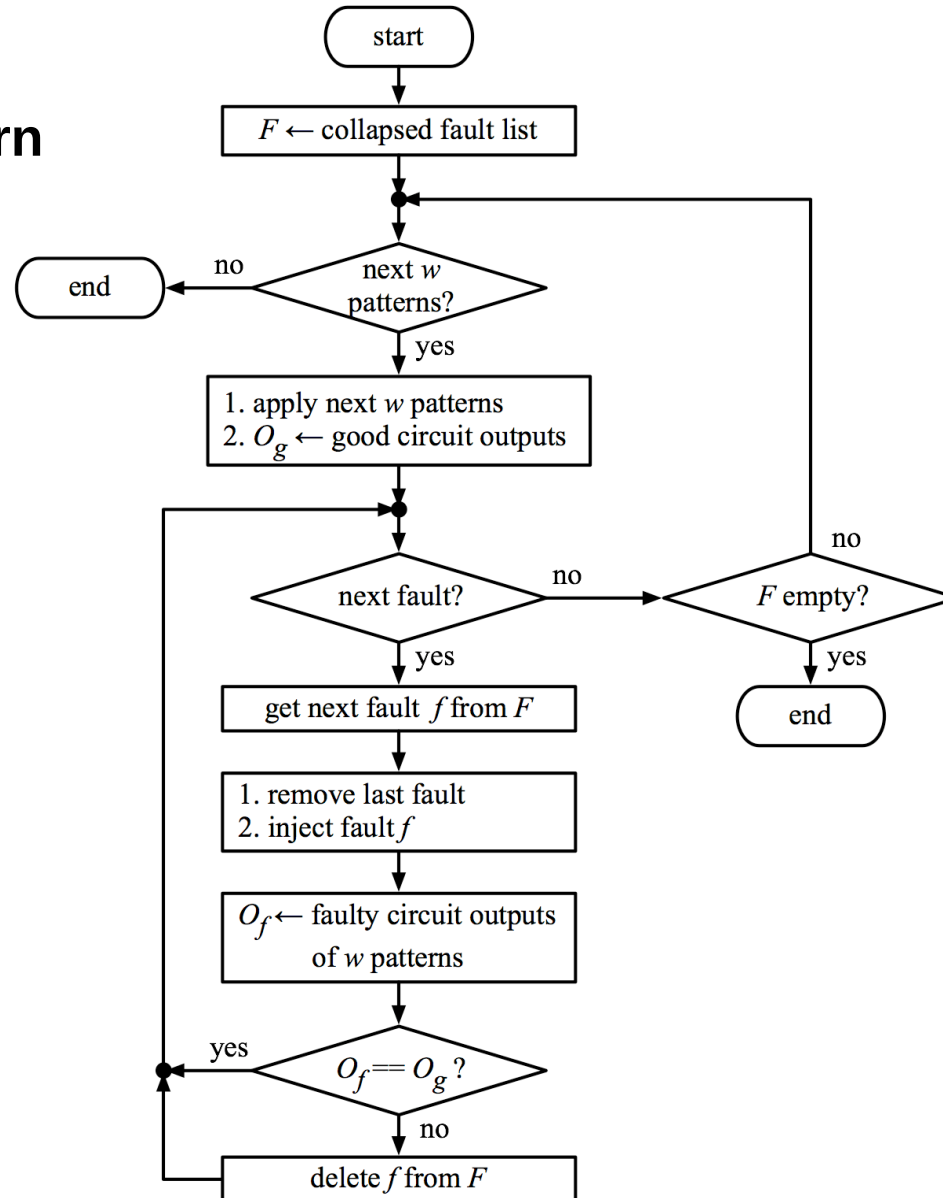
$P_1 P_2$ detect f

P_3 detects g



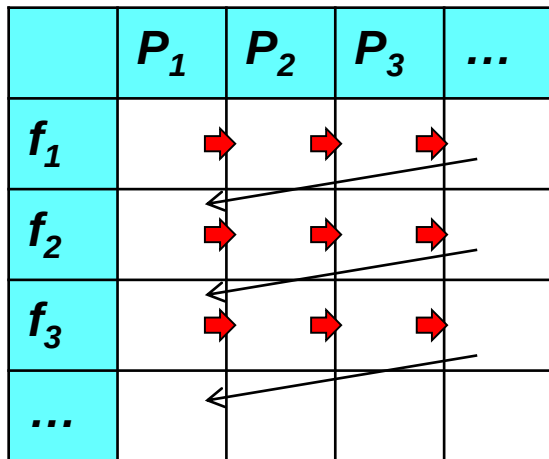
PPSFP Flow Chart

- inner loop: fault
- outer loop: pattern
- (WWW Fig 3.25)

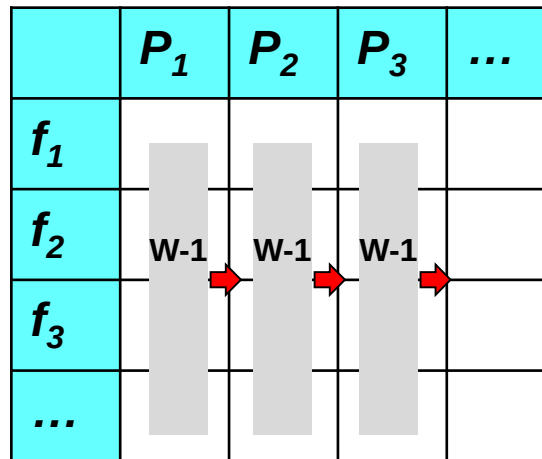


Comparison

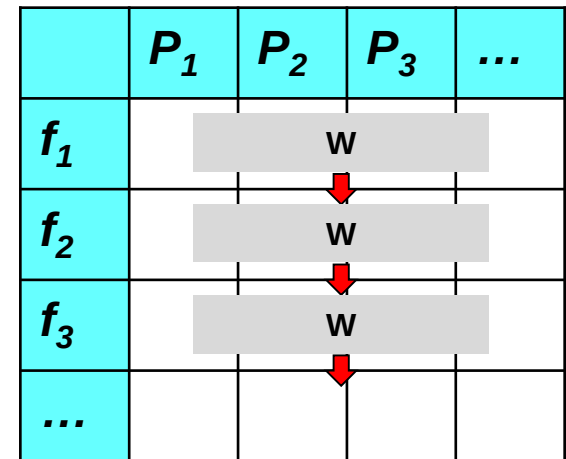
- Serial fault simulation: one fault, one pattern
- Parallel fault simulation: multiple faults, one pattern
- PPSFP: multiples pattern, one fault



- Inner loop: patterns
- Outer loop: faults



- Inner loop: patterns
- Outer loop: faults



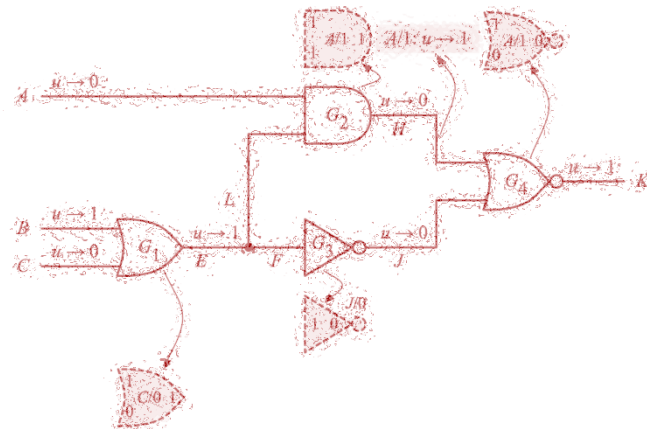
- inner loop: fault
- outer loop: pattern

Parallel pattern assumes combinational circuit

inner loop: event-driven

Summary

- Parallel fault simulation (parallel in fault)
 - ◆ Run parallel logic simulation for $W-1$ faults
 - ◆ Fault injection by circuit modification
- Parallel Pattern Single fault Propagation (parallel in pattern)
 - ◆ Run parallel logic simulation for W patterns



FFT

- Q1: Why parallel fault sim ($W-1$) faults, not W faults?
- Q2: Please compare parallel fault sim and PPSFP
- Q3: PPSFP is useful to comb. ckt only, how about sequential ckt?

	Parallel Fault	PPSFP
Applicable to delay fault model?		
Applicable to Sequential ckt ?		
When to drop fault ?		